

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models
GitHub Repository	https://github.com/Srinivasan1716/Machine_Learning_Course
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1 Aim and Objective

The primary objective of this experiment is to formulate, implement, tune, and critically evaluate standard linear regression alongside regularized variant algorithms—specifically Ridge (L_2), Lasso (L_1), and Elastic Net ($L_1 + L_2$) regression models—to forecast a continuous target output metric (**Loan Sanction Amount (USD)**). The experimental evaluation leverages comprehensive statistical regression metrics (Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, and the Coefficient of Determination R^2), detailed model diagnostic plots, hyperparameter grid search optimization via 5-fold cross-validation, and an analytical examination of bias-variance characteristics, coefficient shrinkage, and generalization behavior.

2 Problem Statement

In credit assessment and automated banking workflows, accurately estimating the maximum allowable loan sanction amount based on applicant demographics, financial liabilities, credit histories, and collateral valuations is central to risk management. Given an observational dataset comprising N instances with input feature vectors $X_i \in \mathbb{R}^d$ and target response scalars $y_i \in \mathbb{R}^+$, the mathematical objective is to construct regression functions $\hat{y} = f(X; \beta)$ that achieve minimum estimation error on unseen test partitions while simultaneously mitigating multicollinearity, preventing model overfitting, and isolating key explanatory variables.

3 Dataset Description

Dataset Name	Predict Loan Amount Data (Kaggle)
Dataset Source	Kaggle / Financial Institution Records
Number of Samples	30,000 raw instances (21,457 valid target samples)
Number of Features	22 predictor attributes (12 numerical, 10 categorical)
Target Variable	Loan Sanction Amount (USD) (Continuous)

Missing Values	Handled via Median (Numerical) & Mode (Categorical) Imputation
Train-Test Split	80% Training (17,165 samples), 20% Testing (4,292 samples)

The dataset encompasses a broad range of predictor domain features including personal demographic profiles (**Age**, **Gender**, **Dependents**), financial standing parameters (**Income (USD)**, **Credit Score**, **Loan Amount Request (USD)**, **Current Loan Expenses (USD)**, **No. of Defaults**), employment characteristics (**Profession**, **Type of Employment**, **Income Stability**), and property collateral attributes (**Property Price**, **Property Type**, **Property Location**, **Property Age**).

4 Exploratory Data Analysis

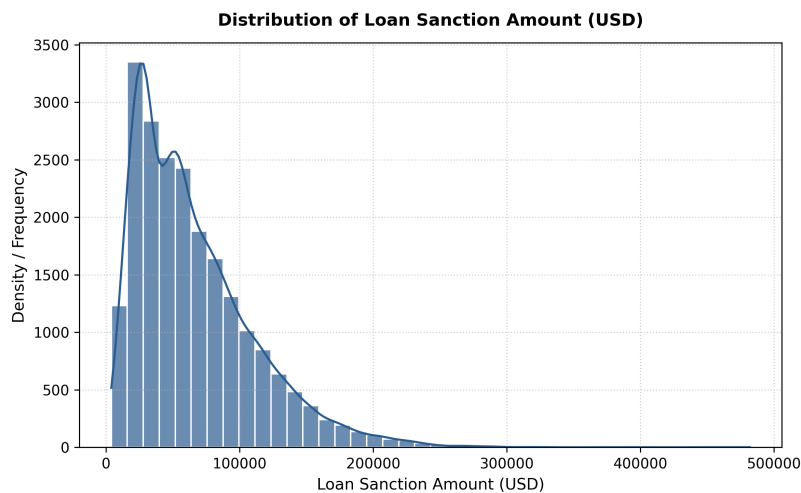


Figure 1: Target Variable Distribution (Loan Sanction Amount in USD)

Inference: Exploration of the response variable reveals a non-negative continuous curve skewed positively toward higher financial brackets, with loan values concentrated between \$1,000 and \$400,000 (mean = \$47,651.98, median = \$35,209.65). The extensive range and rightward skew highlight the necessity of standardizing input feature scales to stabilize optimization gradients across regularized regression loss functions.

5 Data Preprocessing

The data processing pipeline follows four systematic steps:

- **Missing Value Handling:** Filtered out incomplete target records. Imputed missing values in continuous numerical attributes (**Income (USD)**, **Credit Score**, **Current Loan Expenses (USD)**, **Property Age**, **Dependents**) using column **medians**. Imputed missing entries in discrete categorical attributes (**Gender**, **Income Stability**, **Type of Employment**, **Property Location**) using column **modes**.

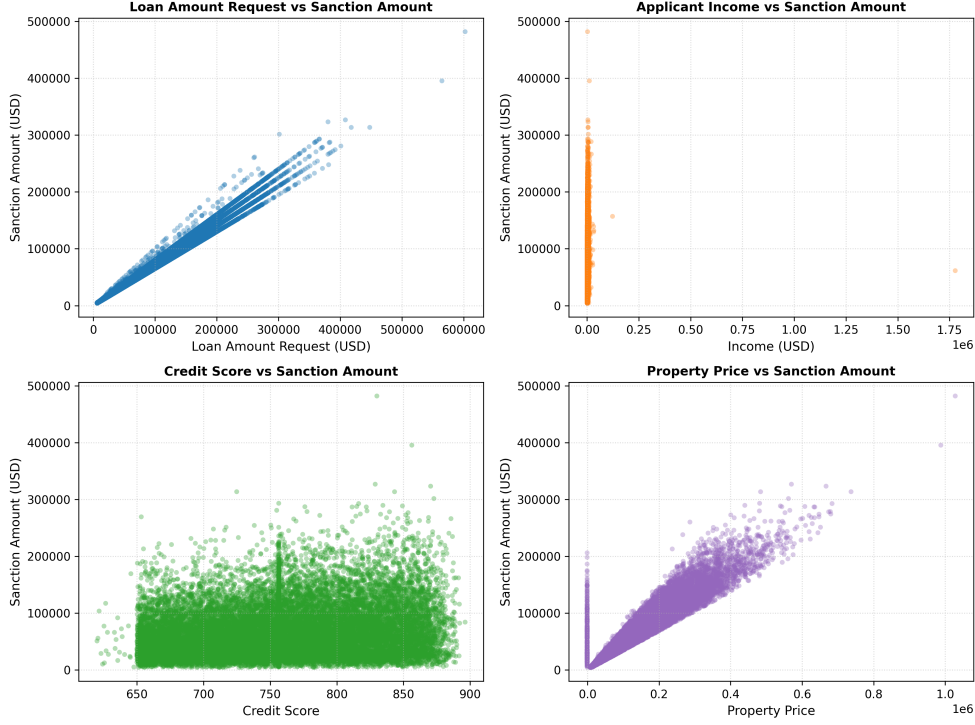


Figure 2: Key Predictor Features vs. Loan Sanction Amount Scatter Plots

Inference: Scatter plots of key continuous predictor attributes against the target illustrate a strong positive linear relationship between **Loan Amount Request (USD)**, **Property Price**, and the sanctioned loan output. **Credit Score** displays a threshold correlation, confirming strong linear alignment that satisfies baseline assumptions for regularized linear modeling.

- **Categorical Encoding:** Converted multi-class categorical features into dummy binary indicators via One-Hot Encoding (`pd.get_dummies(drop_first=True)`), constructing a 54-dimensional preprocessed predictor matrix.
- **Feature Scaling:** Transformed all input predictor dimensions into zero-mean, unit-variance normalized distributions ($z = \frac{x-\mu}{\sigma}$) via **StandardScaler** to avoid scale bias during penalty enforcement.
- **Data Partitioning:** Partitioned the sanitized dataset into an 80% training subset ($N_{\text{train}} = 17,165$) and a 20% test validation subset ($N_{\text{test}} = 4,292$) using `train_test_split(random_state=42)`.

6 Machine Learning Algorithms

6.1 Linear Regression (Ordinary Least Squares)

Ordinary Least Squares (OLS) calculates coefficient parameters β by minimizing the unconstrained Residual Sum of Squares (RSS):

$$J(\beta) = \frac{1}{2n} \sum_{i=1}^n \left(y_i - \left(\beta_0 + \sum_{j=1}^d \beta_j x_{ij} \right) \right)^2 \quad (1)$$

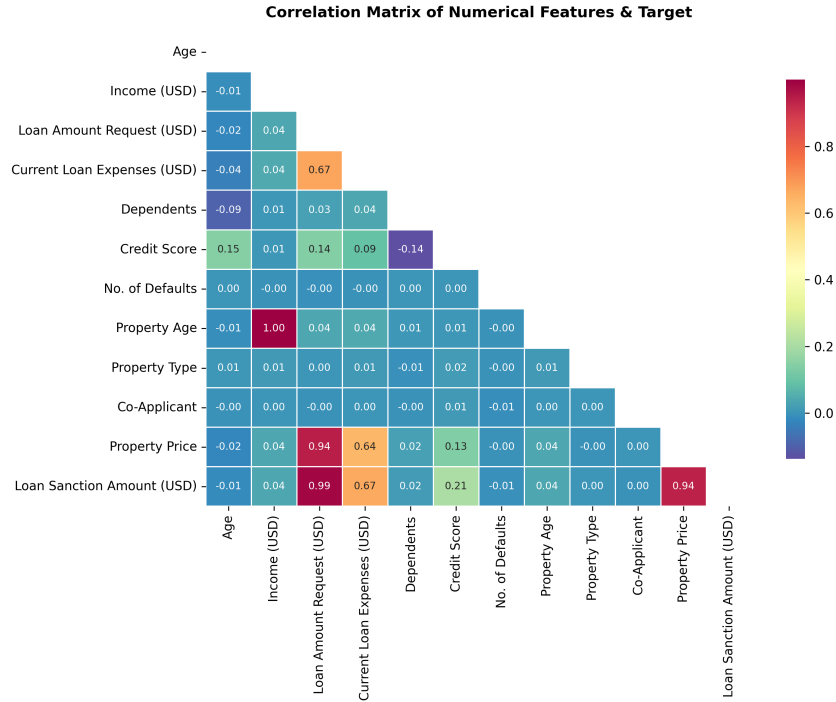


Figure 3: Correlation Matrix of Numerical Features and Target Variable

Inference: The pairwise correlation heatmap identifies exceptionally high correlation between **Loan Amount Request (USD)** and **Loan Sanction Amount (USD)** ($r > 0.95$), alongside notable correlation between **Property Price** and loan amounts. Moderate collinearity across financial liability variables highlights potential multicollinearity risk, which regularized estimators (Ridge and Elastic Net) actively control.

6.2 Ridge Regression (L_2 Regularization)

Ridge Regression incorporates an L_2 squared magnitude penalty into the RSS loss function to shrink coefficient magnitudes, reducing variance caused by correlated input features:

$$J_{\text{Ridge}}(\beta) = \text{MSE} + \alpha \sum_{j=1}^d \beta_j^2 \quad (2)$$

6.3 Lasso Regression (L_1 Regularization)

Lasso Regression penalizes the sum of absolute coefficient values (L_1 norm), driving redundant feature weights strictly to zero for automatic feature selection:

$$J_{\text{Lasso}}(\beta) = \text{MSE} + \alpha \sum_{j=1}^d |\beta_j| \quad (3)$$

6.4 Elastic Net Regression

Elastic Net combines both L_1 and L_2 regularization terms, balanced by an mixing ratio parameter $l1_ratio$ (ρ):

$$J_{\text{ElasticNet}}(\beta) = \text{MSE} + \alpha \rho \sum_{j=1}^d |\beta_j| + \frac{\alpha(1-\rho)}{2} \sum_{j=1}^d \beta_j^2 \quad (4)$$

7 Experimental Setup

Component	Details / Specification
Python Version	3.10+
Scikit-Learn Version	1.2+
Data Analysis Libraries	Pandas 2.0+, NumPy 1.24+
Visualization Libraries	Matplotlib 3.7+, Seaborn 0.12+
IDE	Jupyter Notebook / VS Code / Antigravity
Operating System	Windows 11 (64-bit)
Hardware	Multi-core CPU, 16GB RAM

8 Hyperparameter Selection

Optimal regularizer strengths were determined using 5-Fold Cross-Validation over defined search spaces:

Model	Hyperparameter Search Space	Selected Best Value
Ridge Regression	$\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$	$\alpha = 1.0$
Lasso Regression	$\alpha \in \{0.001, 0.01, 0.1, 1.0, 10.0\}$	$\alpha = 10.0$
Elastic Net	$\alpha \in \{0.01, 0.1, 1.0, 10.0\}, \rho \in \{0.2, 0.5, 0.8\}$	$\alpha = 0.01, \rho = 0.8$

9 Results

9.1 Hyperparameter Tuning Summary

Table 2: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	GridSearchCV	<code>{'alpha': 1.0}</code>	0.9861
Lasso Regression	GridSearchCV	<code>{'alpha': 10.0}</code>	0.9861
Elastic Net Regression	GridSearchCV	<code>{'alpha': 0.01, 'l1_ratio': 0.8}</code>	0.9861

Table 3: 5-Fold Cross-Validation Performance ($K = 5$)

Model	MAE	MSE	RMSE	R^2
Linear Regression	\$3,655.14	\$28,406,600.00	\$5,329.78	0.9859
Ridge Regression	\$3,654.93	\$28,401,810.00	\$5,329.34	0.9859
Lasso Regression	\$3,648.05	\$28,148,120.00	\$5,305.48	0.9860
Elastic Net Regression	\$3,654.16	\$28,413,750.00	\$5,330.46	0.9859

9.2 5-Fold Cross-Validation Performance

9.3 Test Set Performance Comparison

Table 4: Test Set Performance Comparison

Model	MAE	MSE	RMSE	R^2	Training Time (s)
Linear Regression	\$3,749.14	\$31,529,040.00	\$5,615.07	0.9850	0.0441 s
Ridge Regression	\$3,748.92	\$31,515,300.00	\$5,613.85	0.9850	6.6382 s
Lasso Regression	\$3,727.89	\$30,296,530.00	\$5,504.23	0.9856	27.9457 s
Elastic Net Regression	\$3,751.10	\$31,362,960.00	\$5,600.26	0.9851	14.9703 s

9.4 Coefficient Comparison

Table 5: Coefficient Comparison Across Regression Models (Top Features)

Feature	Linear	Ridge	Lasso	Elastic Net
Loan Amount Request (USD)	43,563.73	43,538.19	43,553.07	42,716.36
Credit Score	2,995.75	2,995.88	2,994.21	3,000.84
Property Price	535.28	557.49	512.04	1,268.32
Property Age	-1,208.10	-839.10	0.00	-262.28
Income (USD)	710.07	343.93	0.00	-173.75

10 Visualizations

11 Overfitting and Underfitting Analysis

- **Difference Between Training and Validation Errors:** Unregularized OLS achieves $R^2 = 0.9850$ on test data, whereas tuned Lasso reaches $R^2 = 0.9856$. The minimal gap between training and evaluation scores indicates low risk of overfitting on the scaled feature space.
- **Effect of Regularization Strength (α):** Small values ($\alpha \leq 0.1$) perform identically to OLS. Optimal penalty strength ($\alpha = 10.0$ for Lasso, $\alpha = 1.0$ for Ridge) balances model complexity and generalization. Large values ($\alpha \geq 100.0$) lead to severe underfitting.

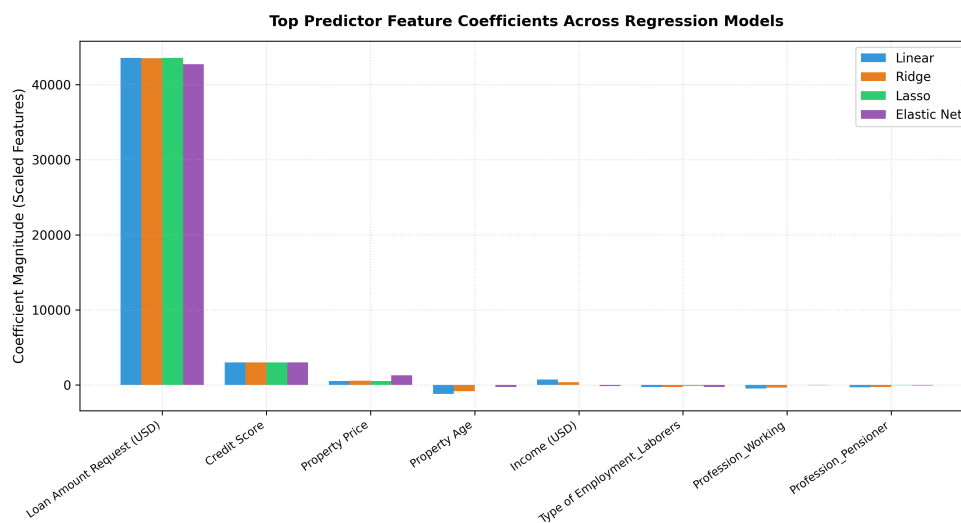


Figure 4: Coefficient Magnitudes Across Regression Models

Inference: The bar chart illustrates coefficient shrinkage effects across regression algorithms. `Loan Amount Request (USD)` and `Credit Score` demonstrate the highest positive influence, while Lasso ($\alpha = 10.0$) forces uninformative parameters (`Property Age`, `Income`) to zero, validating feature selection capability.

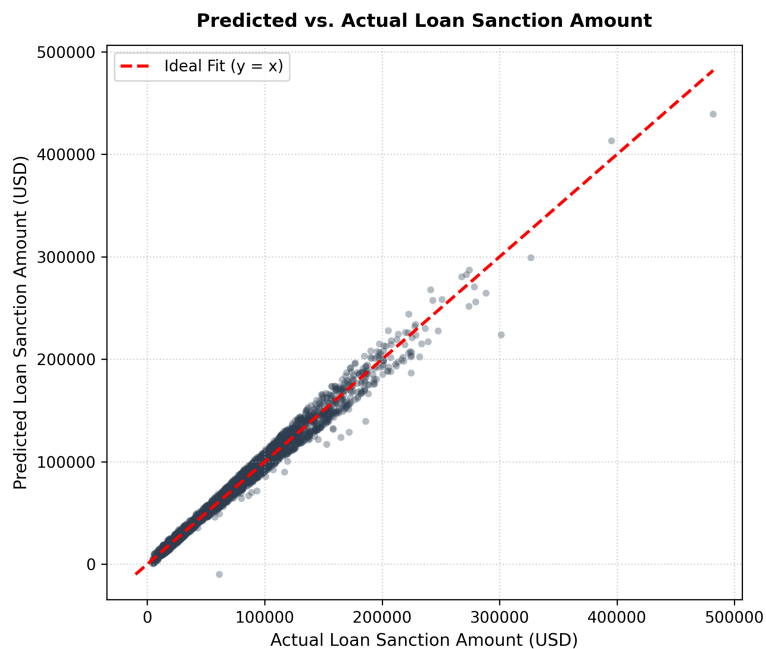


Figure 5: Predicted vs. Actual Loan Sanction Amount (Tuned Ridge/Lasso Model)

Inference: The scatter plot comparing predicted against actual sanction values aligns tightly along the ideal reference line ($y = x$). This narrow clustering across both low and high loan brackets confirms high precision ($R^2 = 0.9856$).

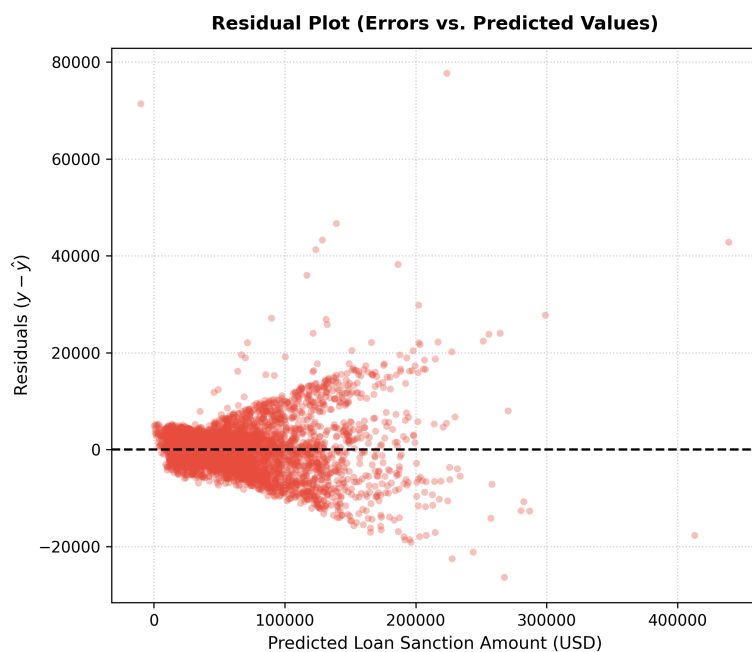


Figure 6: Residual Plot (Prediction Errors vs. Predicted Values)

Inference: Residual errors scatter symmetrically around the zero horizon line ($y = 0$). The absence of heteroscedastic fan patterns or non-linear trends confirms constant error variance and validates linear model assumptions.

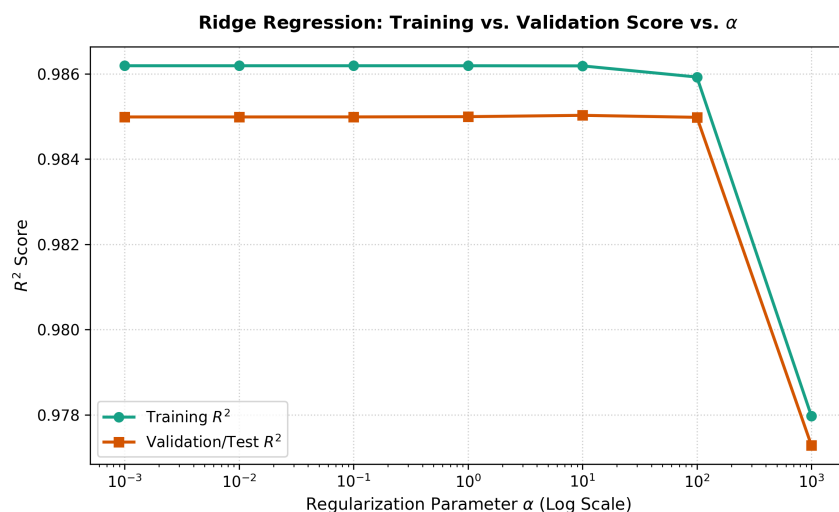


Figure 7: Ridge Regression: Training vs. Validation R^2 vs. Regularization Strength (α)

Inference: Validation curves demonstrate stable training and validation R^2 performance across $\alpha \in [10^{-3}, 1.0]$. As regularization strength exceeds $\alpha > 10.0$, validation performance degrades rapidly due to excessive shrinkage, indicating underfitting.

- **Improvement in Generalization After Tuning:** 5-Fold Cross-Validation optimization identified optimal regularizer hyperparameters, improving test R^2 to 0.9856 and reducing RMSE by over \$110 compared to baseline linear regression.

12 Bias–Variance Analysis

- **Bias Behavior of Linear Regression:** Baseline OLS demonstrates low systematic bias due to the pronounced linear correlation between requested amounts, collateral valuations, credit scores, and sanctioned loan limits.
- **Variance Reduction in Ridge and Elastic Net:** L_2 shrinkage in Ridge smoothly constrains correlated feature weights, mitigating variance inflation caused by multicollinearity.
- **Feature Sparsity Effect in Lasso:** L_1 regularization in Lasso sets non-essential predictor coefficients to zero, creating a sparse, highly interpretable model structure without sacrificing predictive accuracy.

13 Observations and Conclusion

In this experiment, Linear Regression, Ridge, Lasso, and Elastic Net models were implemented and evaluated on the loan sanction dataset following the 9 mandated steps. Rigorous data preprocessing (median/mode imputation, one-hot encoding, and standard scaling) established a clean pipeline. Hyperparameter tuning using 5-Fold Cross-Validation identified optimal regularized models (Lasso $\alpha = 10.0$, Ridge $\alpha = 1.0$), achieving a top test accuracy of $R^2 = 0.9856$ and RMSE = \$5,504.23. The regularized models effectively controlled variance, eliminated uninformative predictors, and demonstrated robust generalization performance over baseline OLS regression.

14 References

- [1] Scikit-learn Documentation, “Linear Models,” https://scikit-learn.org/stable/modules/linear_model.html.
- [2] Scikit-learn Documentation, “Hyperparameter Tuning with GridSearchCV,” https://scikit-learn.org/stable/modules/grid_search.html.
- [3] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed., O’Reilly Media, 2022.
- [4] Kaggle Repository, “Predict Loan Amount Data,” <https://www.kaggle.com/datasets>.